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Importance of secondary inoculum of *Plasmopara viticola* to epidemics of grapevine downy mildew

D. Gobbin[a]*†, M. Jermini[b], B. Loskill[c], I. Pertot[d], M. Raynal[e] and C. Gessler[a]

[a]*Phytomedicine/Pathology, Institute of Plant Sciences Swiss Federal Institute of Technology, Universitätstrasse 2, 8092 Zürich;* [b]*Agroscope, Swiss Federal Research Station for Plant Production of Changins, Centre of Cadenazzo, 6594 Contone, Switzerland;* [c]*State Research Institute Geisenheim, Department of Phytomedicine, E.-von-Lade-Str. 1, 65366 Geisenheim, Germany;* [d]*SafeCrop Centre, Istituto Agrario di San Michele all'Adige, via E. Mach 1, 38010 S. Michele all'Adige (TN), Italy; and* [e]*ITV Aquitaine, Unité de Bordeaux, 39 rue Michel Montaigne, BP 116 Blanquefort Cedex, France*

To quantify the magnitude and the spatial spread of grapevine downy mildew secondary sporangia, 4685 *Plasmopara viticola* single lesion samples were collected from 18 plots spread across central Europe. Disease symptoms were collected on two to 22 sampling dates per plot between 2000 and 2002. Four multiallelic microsatellite markers were used for genotypic identification of pathogen samples. Genetic analysis showed more than 2300 site-specific *P. viticola* genotypes, indicating that populations are genetically rich demographic units. Approximately 70% of the genotypes were sampled once and 14% were sampled twice throughout the various epidemics. In the 18 populations only seven genotypes (0.3%) were identified more than 50 times. Three genotypes particularly successful in causing disease through secondary cycles showed mainly a clustered distribution. The distance of sporangial migration per secondary cycle was less than 20 m and their plot colonization rate was calculated at around 1–2 m^2 day^{-1}. Downy mildew epidemics of grapevine are therefore the result of the interaction of a multitude of genotypes, each causing limited (or a few) lesions, and of a dominant genotype able to spread stepwise at plot-scale. These findings contrast with current theories about grapevine downy mildew epidemiology, which postulate that there is massive vineyard colonization by one genotype and long-distance migration of sporangia.

Keywords: epidemiology, oomycete, population genetics, SSR, *Vitis vinifera*

Introduction

Downy mildew, caused by the diploid, heterothallic (Wong *et al.*, 2001) oomycete *Plasmopara viticola*, is the most important grape (*Vitis vinifera*) disease in temperate climates. *Plasmopara viticola* overwinters as sexually produced oospores in fallen leaves and berries. The current understanding about the epidemiology of the pathogen postulates that the disease starts from a restricted number of germinating oospores that cause primary infections early in the grapevine growing season (Blaeser & Weltzien, 1979; Lafon & Clerjeau, 1988; Schruft & Kassemeyer, 1999). After 5–18 days, depending on the temperature, the sporangia are produced, containing asexually produced zoospores (optimum: 18°C and saturating humidity). The capacity of the oospores to infect is believed to diminish rapidly from May to June, and then abundant secondary infections produced by only a few genotypes are assumed to lead to the explosive increase of the epidemic until leaf fall in autumn. The spatial spread of the disease is commonly believed to rely on the migration of sporangia, which may spread over long distances in a short time (Zachos, 1959; Lafon & Clerjeau, 1988; Blaise *et al.*, 1999).

To predict the risk of secondary or primary infections, weather-driven epidemiological models have been used. Most of these models are based on the above-mentioned assumptions that an epidemic starts from a restricted number of oosporic infections, followed by massive clonal multiplication (Lalancette *et al.*, 1988a,b; Blaise *et al.*, 1999). Two models are based primarily on the simulation of the germination dynamic of oospores. In such models, oospores are considered to be the only factor responsible for the whole epidemic (Hill, 1990, 2000; Strizyk, 1994). However, models often fail to predict the real quantitative development of epidemics or lead to an overestimation of the infection risk, which restricts their use in practice. With the current knowledge available, models cannot be significantly improved without separate quantitative

*To whom correspondence should be addressed.

†E-mail: davide.gobbin@ipw.agrl.ethz.ch

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modelling of sexual and asexual phases (M. Jermini, Agroscope RAC Changins, personal communication).

To gain a better understanding of the magnitude of the primary and secondary cycles of the pathogen, it is important to be able to determine the quantitative contribution of distinct genotypes to a specific epidemic stage. Recently, quantitative investigations of propagule dispersal have been greatly facilitated by the use of genetic markers (Cooke & Lees, 2004), which allow a detailed analysis of population dynamics (Flodgren *et al*., 2000). Polymorphic, specific and co-dominant microsatellite markers (simple sequence repeats, SSRs) have been developed to investigate population genetics of *P. viticola* (Gobbin *et al*., 2003a). Recent studies have shown that *P. viticola* SSRs generally demonstrated a high degree of polymorphism and therefore are useful in discerning genetic variation among individuals (Gobbin *et al*., 2003b; Rumbou & Gessler, 2004).

In the present study, *P. viticola* SSR markers were employed to: (i) determine what proportion of total *P. viticola* genotypes undergoes secondary disease cycles; (ii) quantify the magnitude of secondary sporangia-derived asexual progeny; and (iii) quantify the spatial dispersion of genotypes particularly successful in generating secondary lesions as a function of epidemic progression. If epidemics are triggered by a few oosporic infections and fuelled by a great number of asexual infections, the genetic diversity in the *P. viticola* population structure is expected to be low. Populations should comprise a low number of genotypes, each represented many times.

Grape responses to disease attack and pathogen aggressiveness are subject to considerable variation due especially to variability in macro- and microclimatic conditions. In order to reduce this variability, 18 *P. viticola* epidemics across central Europe were analysed. The value of this study lies in providing quantitative data for a better understanding of the *P. viticola* epidemiological cycle and for improving disease forecasting models.

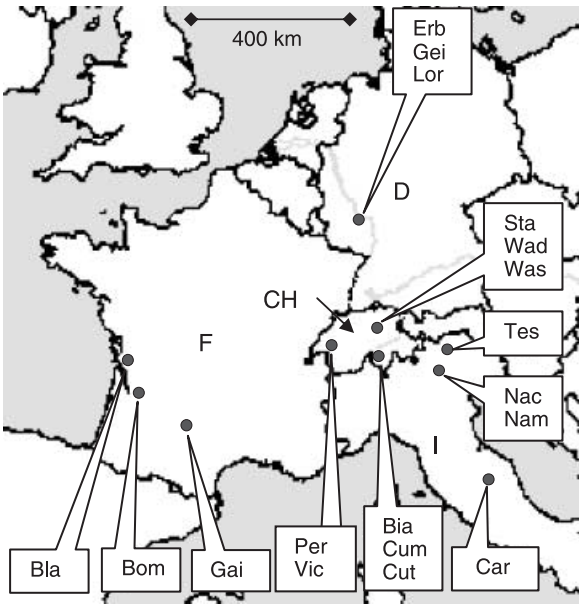


Figure 1 Locations of the 18 downy mildew (*Plasmopara viticola*) populations surveyed in vineyards (*Vitis vinifera*) during the years 2000–02. 1, Biasca (Bia); 2, Blanquefort (Bla); 3, 4, Cugnasco (Cut and Cum); 5, Gaillac (Gai); 6, Navicello (Nam); 7, Perroy (Per); 8, Stäfa (Sta); 9, Wädenswil (Wad); 10, Bommes (Bom); 11, Carpineta (Car); 12, Erbach (Erb); 13, Geisenheim (Gei); 14, Lorch (Lor); 15, Navicello (Nac); 16, Tesero (Tes); 17, Vic (Vic) and 18, Wädenswil (Was).

Materials and methods

Plot locations and epidemics surveyed

Eighteen plots were selected in vineyards in Germany ($n = 3$), France ($n = 3$), Italy ($n = 4$) and Switzerland ($n = 8$), of which 17 were untreated and one, Cum at Cugnasco, Switzerland, was treated four times with fungicides (Fig. 1). Naturally occurring downy mildew epidemics within the 18 plots were surveyed during years 2000, 2001 or 2002. The survey period started as soon as the first lesions (also called oil spots) were noticed and ended when downy mildew generated a mosaic pattern, which impeded the collection of distinct lesions. On average, the survey period lasted 1·5 months and ended in July or August. A sample consisted of 0·5–1 cm^2 of every identifiable lesion (corresponding to approx. one-third to half of the total lesion surface), including up to 0·5 cm^2 of healthy leaf tissue. The remaining part (half to two-thirds) of the lesion was not sampled to ensure the survival of the genotype in the plot. The exact location of each lesion within the plot was recorded according to vine and row number. Oil spots were collected on two to 22 sampling dates per plot throughout the 18 plots; in total, for all plots and all sampling dates, 83 samplings were performed (Table 1). Two different sampling strategies were used: total and partial sampling strategies (TSS and PSS, respectively). The TSS was used when, on average, less than four oil spots per vine were present. Using this strategy, all oil spots were collected as soon as they were visible. A PSS was performed as an alternative to total sampling, when more than four oil spots per vine were present. Using PSS, from one to three oil spots per vine were sampled.

When the TSS was applied on at least two sampling dates within the same plot throughout one epidemic, the total downy mildew population (pooled individuals collected on all sampling dates) was considered to be totally sampled (TSS; Table 1). When the TSS was used on one sampling date (or none, as in Wad) and the PSS on all the remaining sampling dates, the total *P. viticola* population was considered to be partially sampled (PSS, Table 1). Hereafter, TSS or PSS will refer to the sampling strategy applied to the 18 total populations and not to the 83 independent samplings.

In Was, a single vine was infected and all lesions on any sampling date were sampled (TSS) and their positions were recorded according to a reference system defined by six sectors (top, medium, low on both left- and right-hand vine sectors).

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Table 1 Results of sampling grapevine downy mildew (*Plasmopara viticola*) from 18 European populations during the period 2000–02. Plot characteristics (site number, site abbreviation, site name in full and nation), training system, grape variety, number of rows (R), distance between rows (d bet R), number of vines per row (V/R), distance between vines in a row (d bet V), total vines (N V), plot area (A), sampling dates (S date), sampling strategy (SS), sampling size (N_{obs}), number of different genotypes (N_{gen}), number of single genotypes (genotypes that occurred once throughout the survey period; N_{sgen}), name of the dominant genotype (DGEN name), occurrence of the dominant genotype ($N_{obs\ DGEN}$), name of the second dominant genotype (SDGEN name) and occurrence of the SDGEN ($N_{obs\ SDGEN}$). Bold characters refer to the main sampling strategy applied, and to the total number of lesions collected (tN_{obs}), genotypes identified (tN_{gen}), single genotypes identified (tN_{sgen}), DGEN identifications ($tN_{obs\ DGEN}$) and SDGEN identifications ($tN_{obs\ SDGEN}$)

Plot	training syst grape variety	R d bet R	V/R d bet V	N V A[a]	S date	SS[b]	N_{obs}	N_{gen}	N_{sgen}	DGEN name/ $N_{obs\ DGEN}$	SDGEN name/ $N_{obs\ SDGEN}$
										evawasma	*ibevasca*
1-Bia											
Biasca	Guyot	2	21	42	26·5·00	TSS	27	26	18	0	1
Switzerland	Merlot	1·75 m	1·25 m	92 m^2	18·6·00	PSS	96	70	48	5	5
					30·6·00	PSS	101	64	36	12	8
					13·7·00	PSS	90	67	42	3	2
					tot pop	**PSS**	**314**	**190**	**144**	**20 (6·4%)[c]**	**16 (5·1%)[d]**
										ebebissa	*ebetassa*
2-Bla	Double										
Blanquefort	Guyot	4	70	280	10·5·00	TSS	36	30	22	0	2
France	Merlot	1·5 m	1 m	420 m^2	31·5·00	PSS	100	85	53	2	0
					16·6·00	PSS	189	151	94	7	5
					07·7·00	PSS	106	85	45	1	2
					07·8·00	PSS	126	93	62	4	0
					tot pop	**PSS**	**557**	**363**	**276**	**14 (2·5%)**	**9 (1·6%)**
										alovispa	*ibetosna*
3-Cut											
Cugnasco	Guyot	1	6	6	07·6·01	TSS	1	1	1	0	0
Switzerland	Merlot	2 m	1·2 m	14 m^2	25·6·01	PSS	42	17	10	10	14
					23·7·01	PSS	135	56	39	34	21
					tot pop	**PSS**	**178**	**69**	**50**	**44 (23·5%)**	**35 (18·7%)**
										alovispa	*ebebesra*
4-Cum											
Cugnasco	Guyot	4	61	244	07·6·01	TSS	17	9	5	0	6
Switzerland	Merlot	2 m	1·2 m	586 m^2	25·6·01	PSS	41	21	17	0	20
					23·7·01	PSS	340	137	78	45	0
					29·8·01	PSS	86	50	27	10	0
					tot pop	**PSS**	**484**	**193**	**127**	**55 (11·4%)**	**26 (5·4%)**
										abebessa	*aveyusla*
5-Gai	Cordon de										
Gaillac	royat	4	21	79	05·6·00	PSS	21	10	5	10	2
France	Duras	2·5 m	1 m	210 m^2	14·6·00	PSS	26	16	12	9	3
					tot pop	**PSS**	**47**	**22**	**17**	**19 (40%)**	**5 (10·6%)**
										avavissa	*igigomma*
6-Nam											
Navicello	Guyot	5	21	105	22·5·00	TSS	2	2	1	0	0
Italy	Merlot	2 m	1 m	210 m^2	30·5·00	PSS	18	17	12	0	0
					21·6·00	PSS	57	47	40	5	3
					06·7·00	PSS	86	81	74	0	0
					tot pop	**PSS**	**163**	**142**	**127**	**5 (3·0%)**	**3 (1·8%)**
										ebesossa	*efoganta*
7-Per	Guyot										
Perroy	singolo	4	26	88	30·5·00	PSS	85	41	18	17	2
Switzerland	Chasselas	2 m	0·8 m	166 m^2	29·6·00	PSS	131	53	26	26	17
					11·8·00	PSS	109	49	22	19	4
					tot pop	**PSS**	**325**	**108**	**66**	**62 (19·0%)**	**23 (7·1%)**

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Table 1 *Continued*

										evivusme	*abevisme*
8-Sta											
Stäfa	Guyot	4	20	73	27·6·00	PSS	72	44	22	7	11
Switzerland	R × S	2·1 m	0·9 m	134 m^2	26·7·00	PSS	117	65	28	4	9
					18·8·00	PSS	139	60	29	30	11
					tot pop	**PSS**	**328**	**129**	**79**	**41 (12·5%)**	**31 (9·5%)**
										ozewisra	*ebezesma*
9-Wad											
Wädenswil	Guyot	5	3–15	47	21·6·00	PSS	54	21	10	19	6
Switzerland	R × S	1·9 m	1 m	89 m^2	26·7·00	PSS	73	29	17	35	3
					tot pop	**PSS**	**127**	**43**	**27**	**54 (42·5%)**	**9 (7·1%)**
										übetosca	*azowirma*
10-Bom	Double										
Bommes	Guyot	7	40–69	414	< 27·6·01[e]	–	1	1	0	1	0
France	Sémillon	1·5 m	0·9 m	559 m^2	27·6·01	TSS	63	3	1	60	0
					10·7·01	TSS	18	9	2	4	0
					24·7·01	PSS	58	15	5	25	5
					27·8·01	PSS	53	27	20	22	5
					tot pop	**TSS**	**193**	**41**	**28**	**112 (58·0%)**	**10 (5·2%)**
										afevissa	*afowusba*
11-Car											
Carpineta	wild	3	23	57	25·5·01	TSS	18	15	11	1	0
Italy	Sangiovese	2·5 m	2 m	345 m^2	08·6·01	TSS	49	31	17	8	1
					27·8·01	TSS	57	45	31	0	3
					tot pop	**PSS**	**124**	**84**	**59**	**9 (7·3%)**	**4 (3·2%)**
										afolonna	*dgulepma*
12-Erb											
Erbach	Halbbogen	3	12	36	18·5–6·7·00[f]	TSS	230	199	136	0	9
Germany	Riesling	2 m	1·1 m	79 m^2	10·7·00	TSS	11	9	5	2	0
					27·7·00	TSS	1	1	1	0	0
					01·8·00	TSS	45	21	6	7	0
					tot pop	**TSS**	**287**	**192**	**148**	**9 (3·1%)**	**9 (3·1%)**
										egigusra	*jlomaspa*
13-Gei											
Geisenheim	Halbbogen	3	13	39	26·5·00	TSS	7	2	0	3	0
Germany	R × S	1·8 m	1·2 m	84 m^2	31·5·00	TSS	7	2	0	6	0
					05·6·00	TSS	30	8	2	10	4
					19·6·00	TSS	141	25	15	55	31
					06·7·00	TSS	21	8	3	5	5
					tot pop	**TSS**	**206**	**33**	**20**	**79 (38·2%)**	**40 (19·4%)**
										afevimca	*avuzomme*
14-Lor											
Lorch	Rundbogen	3	12	36	05·6·00	TSS	14	9	5	6	0
Germany	Riesling	2 m	1·3 m	93 m^2	06·6·00	TSS	3	3	3	0	0
					16·6·00	TSS	57	33	19	16	0
					19·6·00	TSS	45	38	24	2	0
					20·6·00	TSS	148	107	81	16	4
					21·6·00	TSS	34	22	16	0	13
					27·6·00	TSS	60	49	36	7	1
					tot pop	**TSS**	**361**	**228**	**184**	**47 (13·0%)**	**18 (5·0%)**

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Table 1 *Continued*

										ymatosmo	*abebessa*
15-Nac	Pergola										
Navicello	singola	4	33	132	15·5·00	TSS	6	6	5	0	0
Italy	Chardonnay	2·8 m	0·8 m	296 m^2	22·5·00	TSS	27	19	12	6	2
					30·5·00	TSS	414	351	315	13	6
					tot pop	**TSS**	**447**	**370**	**332**	**19 (4·3%)**	**8 (1·8%)**
										evuzemca	*evivurro*
16-Tes											
Tesero	Guyot	16	18–23	328	23·6·00	TSS	4	4	2	1	1
Italy	Chardonnay	1·5 m	0·8 m	394 m^2	02·7·00	TSS	71	10	6	43	20
					14·7·00	TSS	22	6	3	15	0
					tot pop	**TSS**	**97**	**15**	**11**	**59 (60·8%)**	**21 (21·6%)**
										abedussa	*abegisma*
17-Vic											
Vic	Pinot noir	6	25	136	11·8·00	TSS	74	45	15	8	3
Switzerland	Guyot S	1·65 m	0·8 m	180 m^2	25·8·00	TSS	262	97	40	25	16
					tot pop	**TSS**	**336**	**114**	**55**	**33 (9·8%)**	**19 (5·7%)**
										ezazerma	*emizorma*
18-Was											
Wädenswil	Guyot	1	4	4	01·6·02	TSS	1	1	0	1	0
Switzerland	R × S	1·9 m	1 m	8 m^2	10·6·02	TSS	25	1	0	25	0
					16·6·02	TSS	85	6	4	79	2
					tot pop	**TSS**	**111**	**6**	**4**	**105 (94·6%)**	**2 (1·8%)**

[a]Plot area was calculated as R × V × (d bet r) × (d bet v) (m^2).
[b]TSS, total sampling strategy; PSS, partial sampling strategy.
[c]Percentage refers to the DGEN-derived disease incidence: $DI_{\%\ DGEN}$ ($tN_{obs\ DGEN}/tN_{obs}$).
[d]Percentage refers to the SDGEN-derived disease incidence: $DI_{\%\ SDGEN}$ ($tN_{obs\ SDGEN}/tN_{obs}$).
[e]No sampling was performed prior to 27 June 2001. An 'open' date refers to the putative appearance of the very first primary infection caused by the DGEN *übetosca*.
[f]A summary of the first 19 samplings performed in Erbach is shown. SDGEN *dgulepma* was identified six times on 6 June 2000, once on 9 June 2000 and twice on 6 July 2000.

Sample processing and genotype identification

Automated high-throughput DNA extraction, PCR amplification of the four polymorphic *P. viticola*-specific SSR loci, BER, CES, GOB and ISA, and sequencer-based fragment analysis were used for genotyping the collected samples (Gobbin *et al*., 2003a). Downy mildew genotypes were defined as strains that differed by at least one allele in their SSR profiles and were named with an eight-letter code [formed by merging each letter or letter set according to the SSR allele pattern, e.g. *evawasma* and *ibevasca* identified in Biasca, Table 1]. As SSR mutation rate was estimated between 10^{-2} and 10^{-6} per locus*generation (Schug *et al*., 1997; Brinkmann *et al*., 1998; Schlötterer, 2000) and a maximum of nine generations per genotype may have taken place during the average survey time (1·5 months; Agrios, 1997), SSR loci were assumed to be overwhelmingly stable throughout the survey.

Each oospore was assumed to contain a single nucleus (Burruano, 2000; Gobbin *et al*., 2003a) that was asexually inherited by the progeny (secondary lesions). Therefore, two oil spots showing identical allele pattern (same genotype) were interpreted as clonal progeny, derived from the same oospore. Different genetic profiles of two oil spots were interpreted as being derived from independent oosporic infections. The first appearance of a genotype was considered as an oosporic-derived primary lesion. Any subsequent observation of the same genotype was interpreted as a secondary lesion (Gobbin *et al*., 2003a; Rumbou & Gessler, 2004).

Data and distribution analysis

For each plot, all the individuals collected on the different sampling dates were pooled. The total number of lesions collected (tN_{obs}) and the total number of genotypes (tN_{gen}) were calculated. The number of genotypes that were identified the same number of times ($tN_{obs} = X$) throughout the survey period was determined and the frequency of every occurrence class ($F_{tN_{obs}} = X$) was calculated. The genotypes occurring once throughout the survey period ($tN_{obs} = 1$) were defined as single genotypes. The genotype occurring at the highest frequency was defined as the

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dominant genotype (DGEN), while the genotype occurring at the second highest frequency was defined as the second dominant genotype (SDGEN). In this study, disease incidence was defined as the number of lesions, or observations (N_{obs}), identified in a spatial unit (plot or vine). The contribution of any genotype to the total disease incidence was calculated by dividing the number of lesions generated by the genotype by tN_{obs} and expressed as a percentage ($DI_{\%}$). Scatterplots showing data distribution and DGEN locations in the plots were constructed using the SYSTAT statistical package (Systat, 2000). The same software was used to construct Gaussian bivariate confidence ellipses centred on the means of the spatial coordinates of the lesions ($P = 0{\cdot}68$). Two-sided *t*-tests were performed using a Microsoft Excel spreadsheet (version 2002). In order to evaluate the oil spot distribution pattern, the variance-to-mean ratio (*v/m*) was employed, which gives an indication of the type of distribution (regular, random or clustered) of the data (Campbell & Madden, 1990). As grapevines are regularly planted within plots, they were used as spatial units, or quadrats, to calculate the *v/m* index. For every vine, the cumulative number of lesions caused by the DGEN at a defined date ($cN_{obs\ DGEN}$) or by any other genotype (cN_{obs}) was calculated. Variance and mean variables of $cN_{obs\ DGEN}$ and cN_{obs} per vine were calculated for every plot. The variation of the *v/m* ratio was calculated over time for three plots, Bom (France), Gei (Germany) and Tes (Italy), where clones of DGEN were identified at least 50 times ($tN_{obs\ DGEN} > 50$) and a TSS was performed (Table 1). When *v/m* was < 1, = 1 or > 1, the pattern of distribution was considered regular, random or aggregated, respectively. The Poisson probability distribution was used to test whether a DGEN spatial pattern of distribution was random or nonrandom, based on $cN_{obs\ DGEN}$ per vine. When the frequency of lesion distribution corresponded to the expected frequency of Poisson distribution, determined by the χ^2 test, the spatial distribution was considered to be random. Alternatively, a nonrandom distribution was interpreted either as aggregated or regular (Charest *et al.*, 2002).

Results

The contribution of genotypes to the epidemic

Downy mildew symptoms appeared approximately at the same time period (mid- to end of May) within the sampling years (2000, 2001 and 2002). Exceptions were Bom, Wad, Tes and Sta, where the disease first appeared in June in different years, and Vic, where it appeared in August 2000. In the nine plots from which the total *P. viticola* population was partially sampled (PSS), genetic analysis of 2523 samples was successful, revealing the presence of 1259 genotypes (populations from Bia to Wad, Table 1). From the remaining nine plots where TSS was applied, 2162 samples were analysed and 1083 different genotypes were distinguished (populations from Bom to Was, Table 1). In summary, among the 4685 samples considered, 2342 plot-specific genotypes were identified.

After grouping the genotypes which occurred the same number of times throughout one epidemic ($tN_{obs} = X$) into genotype frequency classes ($F_{tN_{obs}} = X$), single genotypes ($tN_{obs} = 1$) were shown to constitute the greatest percentage of the individuals collected. On average, 71% of all genotypes within the 18 total populations, assessed after both PSS (Fig. 2a) and TSS (Fig. 2b), were identified only once throughout the survey period. The lowest percentage was recorded in Vic, where single genotypes made up 48% of all genotypes; the highest was at Nac, when single genotypes comprised 90% of all genotypes. On average, over the nine populations sampled after PSS, single genotypes were responsible for 37% of the total disease incidence (SD 19%; Fig. 2c). In the other populations (TSS, Fig. 2d), single genotypes caused an average of 31% of the total disease incidence (SD 25%). A high proportion of single genotypes was observed independently from the sampling date (Fig. 3). Even on the last sampling dates, in seven populations (Bia, Bla, Gai, Nam, Car, Lor and Nac) more than 40% of the lesions analysed were new single genotypes.

Genotypes that were identified twice ($tN_{obs} = 2$) in the same plot throughout the season (either twice on the same sampling date or once on two sampling dates) were, on average, 15 and 13% of the total number of genotypes identified, after PSS and after TSS, respectively (Fig. 2a and b). The proportion of disease incidence caused by those genotypes was on average 13·7% (SD 2·2%) after PSS and 11·0% (SD 8·7%) after TSS (Fig. 2c and d).

Based on TSS, 15 of 1083 genotypes (1·4%) were identified more than 10 times, while eight (0·7%) were identified more than 20 times. The same trend was found after PSS, when, out of 1259 genotypes identified, 23 (1·8%) and 11 (0·9%) genotypes were identified more than 10 and more than 20 times, respectively. The populations that included genotypes identified more than 20 times ($tN_{obs} \geq 21$) were: Gei (DGEN 79 times; SDGEN, 40 times), Tes (59 + 21), Per (62 + 23), Cum (55 + 26), Cut (44 + 35) and Sta (41 + 31) (Table 1 and Fig. 3). Seven genotypes that were identified more than 50 times belonged to seven distinct populations.

$F_{tN_{obs}} = X$ and $DI_{\%}$ data distribution within occurrence classes ($tN_{obs} = X$) were equally described independently from the sample collection method (TSS or PSS; Fig. 2). Significant differences ($P > 0{\cdot}05$ after *t*-tests) were never shown when comparing the averages of $F_{tN_{obs}} = X$ and $DI_{\%}$ for $1 \leq tN_{obs} \leq 9$. No comparisons were possible after $tN_{obs} \geq 10$ due to the lack of a sufficient data set.

Downy mildew dispersion patterns

Based on the epidemics studied, *P. viticola* genotypes showed five distinct dispersal patterns: (i) clonal multiplication at distances shorter than 1 m from the putative primary lesion (close to the source) – this pattern was shown in Car (*afevissa*), Cum (*ebebesra*), Gei (*jlomaspa*), Lor (*avuzomme*), Tes (*evivurro*) and Was (*ezazerma*); (ii) clonal multiplication close to the source followed by plot-scaled dispersion (Bia, *ibevasca* and *evawasma*; Bom,

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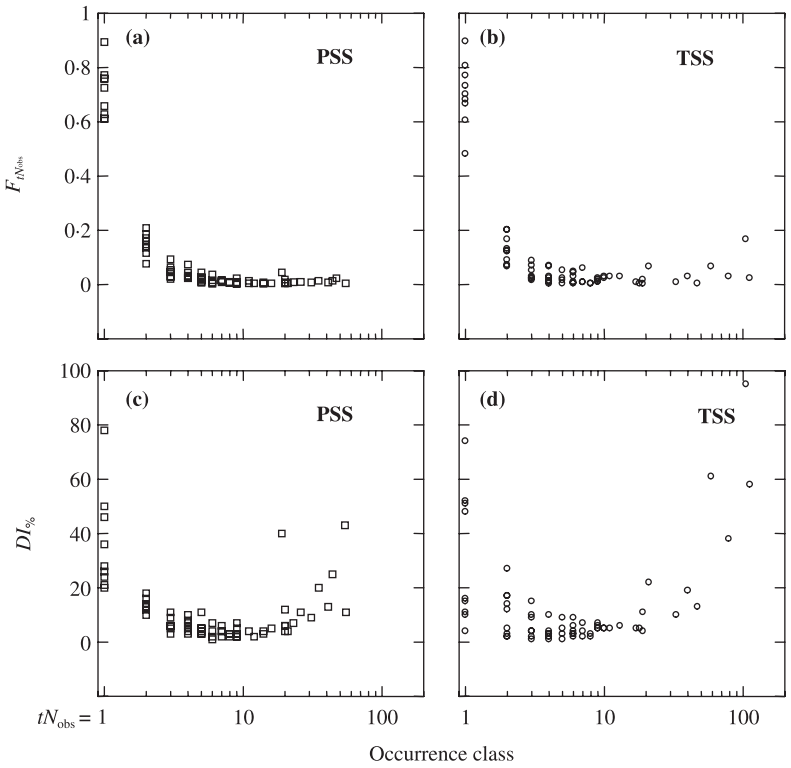


Figure 2 Assessment of frequency ($F_{tN_{obs}} = X$) of *Plasmopara viticola* on *Vitis vinifera* (a, b), and percentage of total disease incidence ($DI_{\%}$) (c, d) of genotype occurrence classes (*x*-axis). The genotypes that were identified the same number of times throughout the survey period were classified into occurrence classes (tN_{obs} = 1, 2, 3 … 112). Square (a, c) and round (b, d) symbols refer to data collected by partial and total sampling strategy (PSS, TSS), respectively. All data points were calculated independently for each of the 18 populations surveyed.

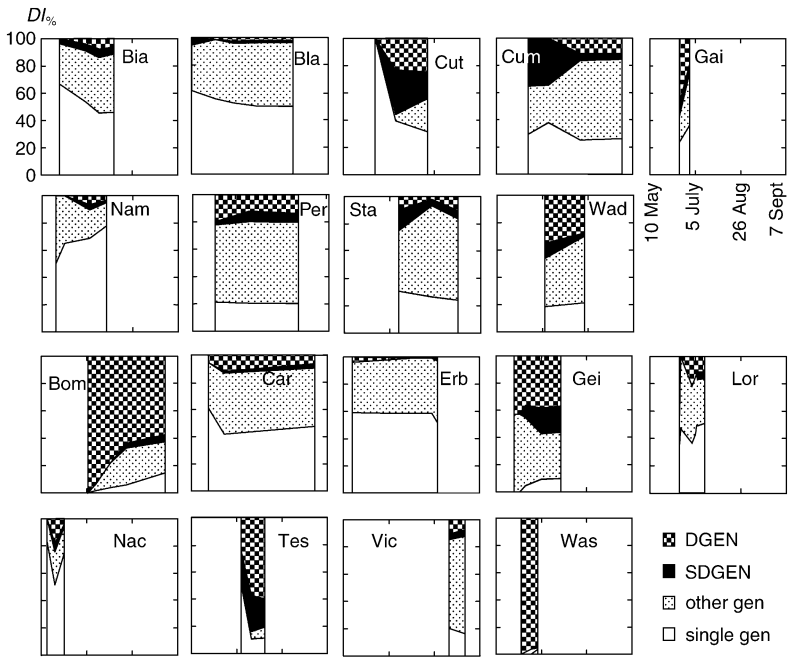


Figure 3 Cumulative proportion of disease incidence (*y*-axis; $DI_{\%}$) caused by the dominant genotype (DGEN) of *Plasmopara viticola* on *Vitis vinifera*, the second dominant genotype (SDGEN), all other genotypes (other gen) and single genotypes as a function of epidemic progress. The survey periods (*x*-axis) were from 10 May to 27 August in the sampling year, either 2000, 2001 or 2002 (Table 1).

übetosca; Cum and Cut, *alovispa*, *ibetosna*; Nac, *ymatosmo*; Per, *efoganta*; Vic, *abegisma*); (iii) multicluster plot-scaled dispersion without previous clonal multiplication close to the source (Lor, *afevimca*; Tes, *evuzemca*); (iv) random plot-scaled dispersion without previous clonal multiplication close to the source (Gei, *egigusra*); and (v) minor clonal multiplication and dispersal. The last dispersal pattern included 99% of the genotypes. The DGEN found in Gai, Per, Sta, Vic and Wad were not strictly classified into any of these first four dispersal patterns because, on the first sampling date, they were already found to be partially plot-scale dispersed. Therefore, due to the lack of a restricted focus, they would eventually be classified into the third or fourth dispersal pattern.

In the following four case studies, four dispersal patterns by genotypes that were identified at least 50 times

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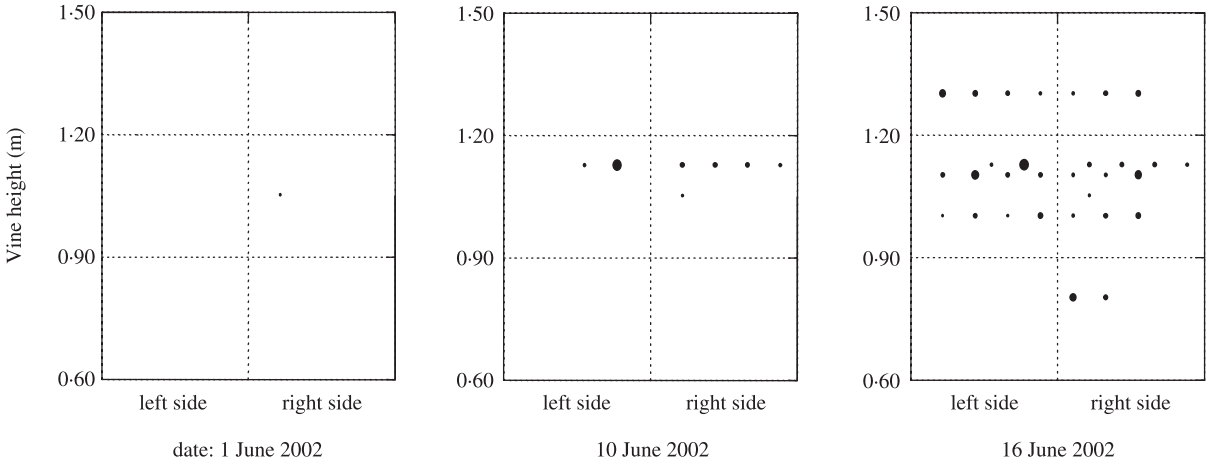


Figure 4 Cumulative spatiotemporal dispersion pattern of the dominant genotype of *Plasmopara viticola* on *Vitis vinifera* in Wädenswil (*ezazerma*). The single vine infected is represented by a bidimensional scaled rectangle, as if it were observed from a neighbouring vine on the same row. Dimensions on the *x*-axis represent left-side and right-side vine extensions, while the *y*-axis represents vine height (m). The grapevine was divided into six sectors (left and right, top, medium and low sectors). Dot size is proportional to the natural logarithm of ($N_{obs\ DGEN}$ + 1) and represents the number of DGEN lesions per infected leaf. The position of infected leaves was assigned randomly within the corresponding sector. Uninfected leaves and leaves infected by other genotypes are not shown. From left to right, plots describe the cumulative genotype spread as a function of time.

each were examined: *ezazerma* (pattern 1), *übetosca* (pattern 2), *egigusra* (pattern 3) and *evuzemca* (pattern 4).

Pattern 1: clonal multiplication close to the source

In Wädenswil (Was) a heavily sporulating oil spot (genotype *ezazerma*) was first detected on 1 June 2002 on a leaf located within the medium-right sector of a vine (Fig. 4 and Table 1). After either one or two asexual cycle(s), observation on 10 June 2002 showed that a spatially restricted infection focus was generated by *ezazerma* in the immediate vicinity of the first infected leaf. A total of 25 new oil spots were detected on six adjacent newly infected leaves: 14 lesions on two leaves within the left-medium sector (12 and two lesions, respectively) and 11 lesions on four leaves within the right-medium sector (three, three, three and two lesions, respectively). On 16 June 2002, presumably after a further secondary cycle, 85 lesions were detected within the same vine. Five new genotypes (six lesions) were identified; one of them (*emizorma*) was already present twice (two clones) on one leaf. Four single genotypes were found on four distinct leaves. *Ezazerma* was found 79 times, colonizing 23 new leaves and sharing two of them with another two genotypes.

Pattern 2: clonal multiplication close to the source followed by plot-scaled dispersion

Among the 64 lesions collected in Bommes 2001, three distinct genotypes were identified. One, *übetosca* (DGEN), was identified 61 times, while the other two were identified once and twice, respectively. At the time of collection, one oil spot out of the 61 lesions successively genotyped as *übetosca* was visibly larger and more necrotized than the remaining 63 lesions. Therefore, it was speculated that this oil spot, sampled on vine 11, row 7 (vine 11/7), may have appeared before 27 June 2001 and that it may have been the putative primary lesion that generated the other 60 lesions by secondary cycles (Figs 3 and 5a; Table 1).

On 27 June 2001, 13 vines (six, five and two vines on rows 6, 7 and 8, respectively) were attacked by *übetosca*: 60% of the lesions were localized on vine 10/7, 13% on vine 11/7, while the remaining 16 lesions were found on 11 vines in the immediate vicinity. Assuming vine 11/7 to be the location of the putative first primary infection (secondary inoculum source), secondary sporangia would have caused infections at an average distance of 1·37 m (SD 1·33). The distance between vine 11/7 and the farther *übetosca* lesion was 6·5 m. During the next 2 weeks, from one to three very mild secondary cycles occurred, which slightly increased the incidence of the DGEN (four new lesions found). The next two samplings revealed that *übetosca* had intensively colonized the northern part of the plot, with 25 and 22 new lesions, respectively. The DGEN *übetosca* had colonized a plot surface of approximately 133 m^2 (area of the confidence ellipse) on 27 August 2001. Roughly, that corresponds to an average plot colonization rate of 2 m^2 day^{-1}.

Throughout the survey period, the *v/m* ratio of the DGEN was always > 0, suggesting an aggregated pattern of distribution (Table 2). From an initial ratio of 20·36 (prior to 27 June 2001) *v/m* decreased as a function of time. On 27 August 2001, it reached the minimum of 7·05, indicating that the DGEN did not succeed in colonizing every grapevine with equal success. The frequencies of the different classes of $cN_{obs\ DGEN}$ did not follow the Poisson pattern of distribution ($P < 0{\cdot}001$), as tested by χ^2. This indicated that the DGEN was never randomly distributed within the plot. Considering all genotypes infecting the vines at any date in Bommes, a clustered pattern of distribution was still detected. When the DGEN was excluded from the population, the χ^2 test indicated

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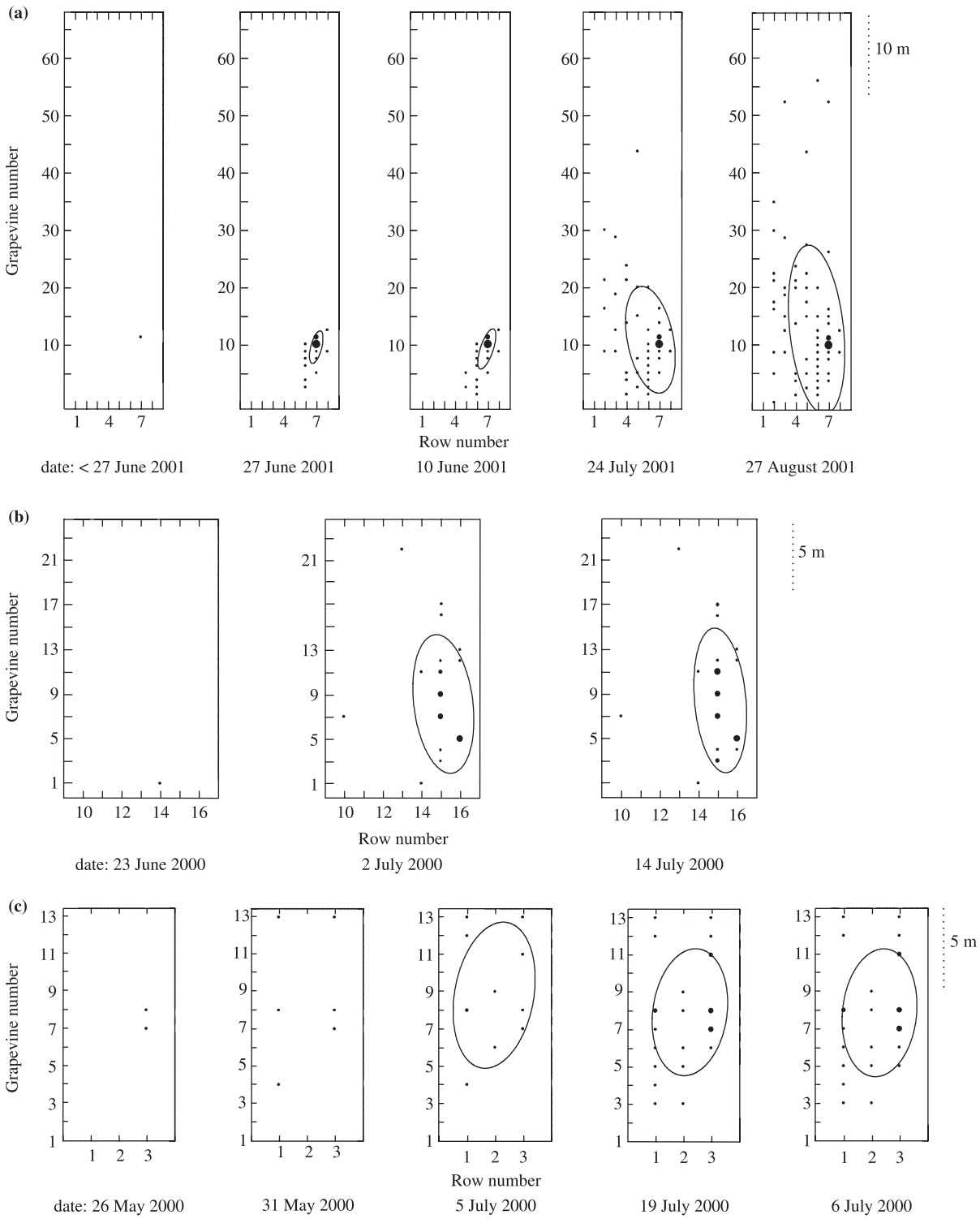


Figure 5 Cumulative spatiotemporal dispersion patterns of the dominant genotypes of *Plasmopara viticola* on *Vitis vinifera*: *übetosca* (a), *evuzemca* (b) and *egigusra* (c) found in Bommes, Tesero and Geisenheim, respectively. The plot is represented by a scaled rectangle (distances are shown by the dotted line on the right-hand side). Vertical *x*-axis ticks describe the beginning and the end of a grapevine row. Numbers on the *x*-axis and the *y*-axis represent grapevine coordinates in the plot (row and grapevine number, respectively). Dot size is proportional to the natural logarithm of ($cN_{\text{obs DGEN}}$ + 1) and represents the cumulative number of DGEN lesions per vine. Gaussian bivariate confidence ellipses, centred on the sample means of *x* and y lesion locations, were specified by a default probability value of 0·68. From left to right, plots describe the genotype spread as a function of time. Row 1 was chemically treated and no sampling was done. The last 14 vines per row are not shown (12·6 m) because they were DGEN-free. In Tesero (b), only the rows showing disease are shown.

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Table 2 Location, sampling date (S date), cumulative number of lesions (cN_{obs}) of *Plasmopara viticola*, cumulative number of infected grapevines (*cNiv*), variance-to-mean ratios (*v*/*m*), calculated for population-specific DGEN, populations without DGEN and SDGEN, and total populations, collected in Bommes (Bom), Geisenheim (Gei) and Tesero (Tes)

Location/ S date	DGEN			Populations without DGEN and SDGEN			Total population		
	$cN_{obs\ DGEN}$ (%)[a]	*cNiv* (%)[b]	*v*/*m*[c]	cN_{obs} (%)	*cNiv* (%)	*v*/*m*	cN_{obs} (%)	*cNiv* (%)	*v*/*m*
Bom	*übetosca*[d]			Removed: *übetosca* [d,e]			All genotypes considered		
< 27·06·01[f]	1 (100)	1 (0·24)	20·36[nc]	0 (0)	0 (0)	nc	1 (100)	1 (0·24)	20·36[nc]
27·06·01	61 (95·31)	13 (3·14)	12·49*[g]	3 (4·69)	3 (0·72)	11·70	64 (100)	15 (3·62)	11·91*
10·07·01	65 (79·27)	16 (3·86)	11·72*	17 (20·73)	17 (4·11)	4·83	82 (100)	31 (7·49)	9·35*
24·07·01	90 (64·29)	39 (9·42)	8·52*	50 (35·71)	49 (11·84)	2·75	140 (100)	81 (19·57)	5·56*
27·08·01	112 (58·03)	58 (14·00)	7·05*	81 (41·97)	77 (18·60)	2·15	193 (100)	126 (30·43)	4·17*
Gei	*egigusra*[d]			Removed: *egigusra*[d] and *jlomaspa*[h]			All genotypes considered		
26·05·00	3 (42·86)	2 (5·12)	4·61	4 (57·14)	4 (10·26)	3·00	7 (100)	4 (10·26)	3·35
31·05·00	9 (64·29)	6 (15·38)	2·76	5 (35·71)	5 (12·82)	2·64	14 (100)	8 (20·51)	2·43
05·06·00	19 (43·12)	10 (25·64)	2·15	21 (47·72)	10 (30·77)	2·62	44 (100)	13 (33·33)	2·11*
19·06·00	74 (40·00)	19 (48·71)	1·78*	76 (41·08)	15 (38·46)	1·94*	185 (100)	22 (56·41)	1·91*
06·07·00	79 (38·35)	20 (51·28)	1·73*	87 (42·23)	20 (51·28)	1·72	206 (100)	24 (61·54)	1·80*
Tes	*evuzemca*[d]			Removed: *evuzemca*[d] and *evivurro*[h]			All genotypes considered		
23·06·00	1 (25)	1 (0·30)	18·11[nc]	2 (50)	2 (0·61)	12·79	4 (100)	4 (1·22)	9·01
02·07·00	44 (58·67)	15 (4·57)	6·85*	10 (13·33)	9 (2·77)	6·20	75 (100)	22 (6·71)	5·18*
14·07·00	59 (60·82)	16 (4·88)	6·24*	17 (17·53)	13 (3·96)	5·45	97 (100)	22 (6·71)	4·74*

nc, statistic not calculated because of a lack of sufficient data points.

*Indicates significance at the 0·001 level.

[a] $cN_{obs\ DGEN}/cN_{obs}$.

[b] *cNiv*/total vines. Total vines refer to 414 (Bom), 39 (Gei) and 328 (Tes).

[c] Variance-to-mean ratio. *v*, variance of cN_{obs} per vine; m, mean of cN_{obs} per vine.

[d] Dominant genotype (DGEN).

[e] Only the DGEN was removed, because the SDGEN caused only 10 lesions.

[f] No sampling was performed prior to 27 June 2001. An 'open' date refers to the putative appearance of the very first primary infection caused by the DGEN *übetosca*.

[g] The Poisson distribution was calculated using $P(x) = [\exp(-m) \times m^x]/x!$. Statistical significance was assessed after χ^2 test.

[h] Second dominant genotype (SDGEN).

a random pattern of distribution ($P < 0{\cdot}05$). Therefore, disease clustering appeared to be an effect of the limited spatial expansion of a sole genotype (DGEN), while the remaining 40 genotypes (Table 1) did not confer any aggregation pattern onto the population.

Pattern 3: multicluster plot-scaled dispersion without previous clonal multiplication close to the source

In Tesero the genotype that contributed the most to the epidemic was *evuzemca* (DGEN): it was identified on vine 1/14 on 23 June 2000 and was responsible for 60% of the disease incidence on 2 July 2000 and 68% on 14 July 2000 (Figs 3 and 5b). On 2 July 2000, 14 vines were infected, mostly vines 5/16 (11 lesions on nine leaves), 7/15 (eight lesions on seven leaves) and 9/15 (eight lesions on seven leaves). The other 16 lesions were dispersed over 11 vines. Considering vine 1/14 as the secondary inoculum source, the average dispersion distance was 6·41 m (SD 2·87; max. 16·88; min. 2·19). A weaker dispersion occurred in the period 2–14 July 2000: the DGEN was identified 15 times, mostly on the vines that were already colonized (only vine 4/16 was newly colonized by a DGEN clone). With an average colonization rate of approximately 1·3 m^2 day^{-1}, the DGEN invaded only 7% of the total plot area over a 3-week period. The SDGEN (*evivurro*) caused 20 lesions at the beginning of July (28% incidence), but afterwards it was not identified again. It followed pattern 1 (see above and Gobbin *et al*., 2003b).

Considering the total population, the *v*/*m* ratio decreased from 9·01 to 4·74 throughout the epidemic. On the second and third sampling dates, the aggregation index was significant when DGEN alone or total populations (including DGEN) were considered, indicating a nonrandom pattern of symptom distribution. After removing *evivurro* and *evuzemca* from the population, significance was no longer achieved ($P < 0{\cdot}05$; Table 2).

Pattern 4: random plot-scaled dispersion without previous clonal multiplication close to the source

In Geisenheim on 26 May 2000, clones of the DGEN *egigusra* were found colonizing two vines (Fig. 5c and Table 1). One lesion was found on vine 7/3 and two on vine 8/3. At least one secondary cycle had already occurred and the first primary infection from which the three DGEN clones had originated was missed by the observers. On 31 May 2000, *egigusra* was identified on four vines. Assuming that vine 8/3 carried the source of the inoculum, the average migration distance was 4·97 m

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with a SD of 1·54 m (max. 7·00; min. 3·6). It took 10 days from the date of the first appearance to colonize 10 vines (26%) dispersed throughout the plot. One disease focus was observed on the fourth sampling date on the two vines where the DGEN was first observed: 13 new oil spots were located on vine 7/3 and 11 new oil spots on vine 8/3. Forty days after the first observation, on 6 July 2000, lesions were observed on 20 vines (51%). With a colonization rate of about 0·8 m^2 day^{-1}, the DGEN was found on vines covering 38% of the total plot surface. On 5 June 2000, four clones of the SDGEN (*jlomaspa*) were identified on vine 12/3, and on 19 June 2000, clones were found on three vines (once on 8/2, seven times on 11/3 and 22 times on 12/3). The secondary sporangia migration distances spanned from 0·01 to 5·13 m (mean 0·44; SD 1). On 6 July 2000, five outbreaks were identified in the vicinity of older infections (occurring three times on vine 12/3 and twice on vine 13/3). The contributions of the DGEN and the SDGEN to the total disease incidence are shown in Fig. 3.

The *v/m* ratio of the DGEN decreased from 4·61 (26 May 2000) to 1·73 (6 July 2000). The χ^2 test suggested a nonrandom pattern of distribution as soon as the focus localized on vines 7–8/3 appeared (19 June 2000, Table 2). *v/m* ratios (always > 1) did not differ substantially when considering spatial distribution of DGEN, populations without DGEN and SDGEN, and total populations. This indicated that clustering, when present, was not strictly caused by *egigusra* and *jlomaspa*, but by the other genotypes as well.

Discussion

In this study, two kinds of sampling strategy were applied: PSS and TSS. The sampling strategies chosen were the response to the progress of the epidemics. The TSS was feasible only when fewer than 1000 lesions were generated throughout the survey period (plots 10–18 in Table 1). Such epidemics are not typical of central Europe, where a mild, rainy climate favours explosive disease development. In the remaining nine epidemics a PSS was employed, allowing a collection of one to three lesions from every vine in the plot. The main difference between sampling strategies is related to their ability to find highly aggregated clones. Partial sampling failed to detect clones restricted to an area smaller than the resolution of the grid and did not allow exhaustive sampling within single vines. As a result, this strategy underestimated the number of progeny per genotype (Fig. 2a and b, right side). Nevertheless, differences among data collected by both sampling types were nonsignificant: PSS and TSS equally determined the frequency ($F_{tN_{obs}} = X$) and the disease incidence ($DI_{\%}$) of every occurrence class ($tN_{obs} = X$).

This study provides a wide-scale quantitative analysis of the proportion of *P. viticola* genotypes that undergoes secondary multiplication and of the genotype-specific distribution within plots during an epidemic. One of the most salient results of this study is that the findings challenge the generally accepted view that an epidemic starts from a restricted number of primary infections, followed by massive asexual reproduction. Instead, there was a continuous input of new genotypes into the epidemic. This phenomenon also occurred under very dry conditions in the Greek population of Aghialos (60·2 mm rain from 6 May to 10 July 2001, and 12·6 mm rain from 8 May to 2 July 2002): on each of the nine sampling dates, between five and 30 new genotypes were identified within the samples collected (Rumbou & Gessler, 2004).

About three-quarters of the genotypes sampled, depending on the epidemic and sampling strategy, never gave rise to asexual progeny. The overall failure to generate secondary lesions strongly contrasts with the high potential for sporangia production by a sporulating lesion. Reuveni (2003) estimated that about $2{\cdot}4 \times 10^5$ *P. viticola* secondary sporangia are produced on 1 cm^2 of an artificially infected vine. Even if produced in such a high quantity, only a small number of them apparently succeed in infecting vines under field conditions. The reasons for this failure may be linked to a high sensitivity to environmental conditions, such as long-wavelength UV light, as was shown for other downy mildews (Rotem *et al.*, 1985). Furthermore, leaf wetness durations and temperatures that deviate from the optimum may be other factors that negatively affect their longevity.

About one-quarter of the genotypes sampled underwent asexual cycles: their clones were identified two to 20 times throughout the survey. A robust quantitative description of the distribution patterns of these genotypes was impeded by the low number of clones generated, and therefore distribution analysis was focused on four of the seven genotypes that were identified more than 50 times. The different dispersion strategies found were the result of the interaction of climatic conditions, cultivation systems, host variety and strain-specific aptitude in generating secondary lesions. In Was (*ezazerma*) and Tes (*evivurro*), but also in Car (*afevissa*) and Cum (*ebebesra*, results not shown), the genotypes intensively colonized a single vine but did not spread to other vines. This initial spatially limited aggregation of secondary inoculum (distance among clones < 1 m) seems to be the antecedent step to the genotype dispersing further (pattern 2); this was observed, for instance, in Bom (*übetosca*) and in Cugnasco (Cut and Cum plots; *alovispa* and *ibetosna*). In the Cugnasco case study, the DGEN and SDGEN first originated in the Cut plot, increased their number on a few vines after secondary cycles, and finally migrated to the Cum plot located 10 m away (results not shown). Further dispersal patterns followed the examples of Tesero (*evuzemca*), where secondary sporangia generated clusters, and Geisenheim (*egigusra*), where sporangia were dispersed throughout the plot generating a random pattern.

The degree of disease aggregation was high at the beginning of the epidemic and decreased as the disease progressed. The highest value was calculated for *übetosca* in Bom on the first sampling date (*v/m* = 20·36). Significant aggregation of the disease symptoms was detected exclusively when considering the dispersion pattern of the DGEN or the total population. The opposite result was

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obtained when the DGEN and SDGEN were removed from the populations. Thus downy mildew populations showing a clustered character may hide one or two dominant genotypes, which are multiplying asexually. Conversely, a random pattern distribution may be interpreted as the absence of a dominant genotype or the presence of nondominant genotypes.

In Bom, Car, Cum, Gei, Lor and Was, but not in Tes, a strong aggregation of secondary lesions was found in the proximity of the putative primary lesion (source of sporangia). The amount of diseased plant tissue decreased rapidly as the distance from the source increased. As suggested by Aylor (1999), the diminishing of disease incidence is due to a rapid decrease of the aerial fungal spore concentration and deposition on plants with increasing distance from the source. This was shown by dispersion tests aiming to quantify spatial dispersal of airborne propagules. Skogsmyr (1994) examined the occurrence of pollen-mediated gene flow from transgenic to nontransgenic potato plants. Recipient plants placed at 0–1 m and 1–2 m from donor plants showed 72 and 32% pollination, respectively. In a similar study, Messeguer *et al.* (2001) found that recipient rice plants placed 1 m away from the donor plants were 50 times more pollinated that recipient plants placed at a distance of 5 m. Following these studies, an aggregation of clones indicates the presence of the primary lesion in the immediate surroundings and rules out the possibility of propagules immigrating across a given distance.

Genotypes of *P. viticola* were found to colonize plot areas at a relatively slow rate, roughly calculated at ~1–2 m^2 day^{-1}. This slow rate of movement may have been due to the spatially discontinuous distribution of hosts. Corridors between rows, roads, woods and rivers act as natural barriers against a homogeneous and omnidirectional disease spread. A second contributory factor may be the weather conditions occurring at the moment of sporulation. Sporangia of *P. viticola* primarily become airborne in the presence of rain or dew (Agrios, 1997). As suggested by Aylor & Sutton (1992) in the case of ascospore release by *Venturia inaequalis*, rainfall reduces the dilution of the spore cloud by turbulence. This is due to the fact that raindrops sweep through the spore-filled column of air and transport spores to the ground, reducing the likelihood of the long-range migration of sporangia.

As a consequence of an infrequent long-range migration, genotypes are rarely shared between neighbouring plots, as supported by three of our case studies. In Navicello, the surveyed plots located 50 m apart (Nac and Nam) shared one genotype among 370 (in Nac) and among 142 (in Nam). Similarly, Gobbin *et al.* (2003b) reported that in San Michele, plots located 500 m apart did not share any of the 171 genotypes identified in total. In Cugnasco, two genotypes (*alovispa* and *ibetosna*) were shared between Cut (69 genotypes) and Cum (193 genotypes), which were located 10 m apart (in Cum, *ibetosna* was identified 20 times; results not shown).

The findings of this study consistently contradict current assumptions in viticulture. The long-range secondary sporangia migration has been overestimated in the past, while the contribution of oosporic infections to an epidemic has gone largely unrecognized. Massive asexual reproduction seems to be the feature of a restricted number of genotypes, while the great majority of them each contribute to a limited proportion of the total disease incidence.

It is now understandable why disease-forecasting models based purely on secondary cycles (Blaise *et al.*, 1999) or solely on subsequent oosporic infections (Hill, 1990) unexpectedly fail. While the former ignore the continuous occurrence of oosporic infections throughout the epidemic, the latter ignore the fact that 20–90% of the disease is caused by one or two dominant and asexually multiplying genotypes. A new concept for an epidemiological model should be based on the integration of two different algorithms, each one dealing with a downy mildew cycle.

If secondary inoculum plays a minor role in pathogen dissemination, it is still unclear how downy mildew spread rapidly across Europe after its introduction in 1878. Oospore-mediated gene flow may be more efficient, because of the long viability of oospores (at least 5 years), which would permit more extensive spread of the disease (G. K. Hill, SLVA Oppenheim, personal communication). Alternative explanations may rely on multiple introduction events followed by human-mediated transportation of diseased plants (rooted grape cuttings carrying soil that contains oospores) or on pathogen introduction(s) prior to 1878.

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